Research Notebook

CSEG3060 • Research Methodology in Computer Science

Your Full Name

SAP ID	500XXXXXX
Batch	B1, B2, B3
Track	Full Stack AI
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B.Tech. Computer Science and Engineering • Final Year

School of Computer Science • UPES

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Declaration. This notebook records my own understanding and progress. Where I have discussed ideas with classmates or worked in a group during class activities, the entries below are still written in my own words. All sources are cited, and any assistance is disclosed in Section 13.

How to use this notebook

This notebook has **fifteen sections**, in a fixed order. They are not fifteen separate assignments — they are one research idea, developed step by step. By the end of the semester the notebook should answer:

> *What is my research problem, why does it matter, what has already been done, what is still missing, how will I investigate it, how will I evaluate it, and what contribution do I expect to make?*

The four rules

1. **Maintain it individually.** Group discussion is encouraged; the notebook is yours.
2. **Add something after every lecture.** At least one meaningful entry, every class. Record it in the log below.
3. **Show the evolution of your thinking.** Do not overwrite. Use `\version{2}{date}{why}` to stamp a revision and keep the earlier text, or `\wasnow{old}{new}` to show a phrase changing. Changes are evidence of research, not of error.
4. **Use credible sources.** Every paper goes into `references.bib` with enough detail to find it again.

What is in each section

- A blue **WHAT THIS SECTION IS FOR** box — read it first.
- An amber **COMMON TRAP** box — the mistake most students make here.
- A grey **WORKED EXAMPLE** box — a complete example, carried through all fifteen sections so you can watch one idea develop. **Delete each grey box once you have written your own.**
- A black-framed **YOUR WORK** box — where you write.
- A teal **DONE WHEN** box — the checklist to test yourself against.

Useful commands

Command	What it does
`\fillme{a one-sentence question}`	Grey italic placeholder. Replace it.
`\version{2}{2026-09-03}{after I.6}`	Stamps a revision, so the evolution stays visible.
`\wasnow{old wording}{new wording}`	Strikes the old wording, shows the new.
`\rnsub{A sub-heading}`	A heading inside a box.
`\todo{find the dataset licence}`	An amber reminder to yourself.
`\citet{key}` / `\cite{key}`	Cites a paper from `references.bib`.

Lecture log

One line after every class. This is the fastest way for both of us to see that the notebook is being maintained continuously rather than assembled the night before. *(If you would rather not keep a log, delete this subsection — nothing else depends on it.)*

Date	Lecture	What I added or changed
[2026-08-04]	L0	*[listed three areas that interest me]*

Contents

1 Research Interest

Lecture I.1–I.2 Update this section after these lectures.

WHAT THIS SECTION IS FOR

Name the areas of computer science you are actually willing to spend two semesters inside. This is the widest section of the notebook and the only one you are allowed to be vague in — everything after it narrows.

Write **three** candidate areas, not one. For each, answer in one line each: what draws you to it, what you have already built or read in it, and who outside this classroom works on it. An area you cannot name a single practitioner or paper for is a preference, not a research interest yet.

COMMON TRAP

Method-first framing. “I want to use transformers / Kubernetes — now let me find a problem.” You will force a method onto a problem that never needed it, and a reviewer will ask why. Choose the area by the *questions* it asks, not the tools it uses.

WORKED EXAMPLE — read it, then delete this whole box

Area 1: Retrieval-augmented generation for closed-domain question answering. Why: I built a chatbot over our department FAQ for a mini-project and it confidently invented a fee deadline. Already done: a LangChain + FAISS prototype; read the original RAG paper Lewis et al. [2020]. Who works on this: FAIR, AI2 (Asai et al.), ACL and EMNLP.

Area 2: Evaluation of generative systems. Why: I could not tell whether my prototype was good, and that bothered me more than the bug. Already done: read the hallucination survey Ji et al. [2023] and skimmed RAGAS Es et al. [2024]. Who works on this: the LLM-evaluation community; ACL, NeurIPS Datasets & Benchmarks.

Area 3: ML systems engineering — serving, latency, cost. Why: the model that wins on a leaderboard is not the model I can serve in 2 seconds. Already done: FastAPI deployment; read the technical-debt paper Sculley et al. [2015]. Who works on this: MLSys, industry engineering blogs.

Leading choice, for now: Area 1 with Area 2 as the lens — I have the prototype and the documents already, which makes feasibility survivable, and Area 2 gives me the metric I was missing.

YOUR WORK

Area 1: *[name the area]*
Why it interests me. *[2–3 sentences, specific]*
What I have already done here. *[a course, a mini-project, a paper you read]*
Who works on this. *[one lab, company, or author, and one venue]*

Area 2: *[name the area]*
Why it interests me. *[2–3 sentences]*
What I have already done here. *[. . .]*
Who works on this. *[. . .]*

Area 3: *[name the area]*

Why it interests me. *[2–3 sentences]*
What I have already done here. *[. . .]*
Who works on this. *[. . .]*

My leading choice, for now

[which of the three, and the single reason]
This may change. If it does, say so here rather than deleting the old text.

DONE WHEN

Three areas named. Each has a named venue, lab or author attached. One is marked as the current front-runner, with a reason.

2 Problem Ideas

Lecture I.2–I.3 Update this section after these lectures.

WHAT THIS SECTION IS FOR

Turn the leading area into **five candidate problems**. Problem-finding is a search, not a spark — run the five-source sweep from Lecture I.2 and record which source produced each idea:

- **S1. Literature gap** — explicit (Future Work), population, contradiction, or assumption.
- **S2. Professional practice** — your own measurable irritation.
- **S3. Replication** — a published result nobody has checked.
- **S4. Interdisciplinary overlap** — CS × another field.
- **S5. Industry pain** — issue trackers, post-mortems, engineering blogs.

Ideas sitting in *two* sources are usually strongest: one source supplies the novelty, the other supplies the significance.

COMMON TRAP

"No papers exist, so this is a good topic." Sometimes nobody asked because the answer is obvious or useless. Absence of recent literature usually means *closed*, not open. Check before you celebrate.

WORKED EXAMPLE — read it, then delete this whole box

#	Candidate idea	Source	Why it might matter	Keep?
1	Unsupported answers in RAG over institutional documents	S2 + S1	My prototype does it; published evals use clean open-domain corpora	Y
2	Does chunk size interact with "lost in the middle" on long policy documents?	S1	Explicit population gap; cheap to test	Y
3	Reproduce RAGAS faithfulness scores against human judgement on my corpus	S3	Metric-validity question; needs no new system	Y
4	Multilingual RAG for Hindi-language circulars	S4	Real and local, but I cannot get enough Hindi documents cleared in two weeks	N
5	Fine-tune a 7B model on university data	S?	Method-first — I wanted to fine-tune something and worked backwards	N

Taking forward: (1) unsupported answers under a latency budget; (2) chunk size × position effects; (3) faithfulness-metric validity.

Rejected: idea 4 — document clearance is outside my control and outside two weeks. Idea 5 — I caught myself doing method-first framing; nothing changes for anyone if it works.

YOUR WORK

Record every idea you generate, including the ones you reject — rejected ideas with a stated reason are evidence of judgement, and they are marked.

#	Candidate idea	Source	Why it might matter	Keep?
1	*[. . .]*	S*[. . .]*	*[. . .]*	*[Y/N]*
2	*[. . .]*	S*[. . .]*	*[. . .]*	*[Y/N]*
3	*[. . .]*	S*[. . .]*	*[. . .]*	*[Y/N]*
4	*[. . .]*	S*[. . .]*	*[. . .]*	*[Y/N]*
5	*[. . .]*	S*[. . .]*	*[. . .]*	*[Y/N]*

The three I am taking forward

1. *[idea, one sentence]*
2. *[idea, one sentence]*
3. *[idea, one sentence]*

Rejected, and why

[one line per rejected idea — "needs hospital data I cannot get", "already solved in 2019", "I could not name a metric"]

DONE WHEN

Five ideas recorded with their source number. At least one comes from source 2 (your own practice). Three carried forward, and every rejection has a stated reason.

3 Problem Statement

Lecture I.3–I.5 Update this section after these lectures.

WHAT THIS SECTION IS FOR

One problem, narrowed until roughly **one** final-year project would satisfy the sentence. Use the funnel — *interest* → *field* → *topic* → *task* → *problem* → *question* — and turn about three of the five knobs: **population, task, comparison, metric, setting**.
Then write it in one of the five templates:

- **T1 Comparative** — "Does X improve *metric* over *baseline*, on *data*, under *constraint*?" (the default)
- **T2 Transfer** — "To what extent does a method validated on P_1 retain *metric* on P_2?"
- **T3 Explanatory** — "Why does *method* fail on *cases*, and which factor accounts for it?"
- **T4 Feasibility** — "Can a system achieve *outcome* subject to C_1 and C_2, and at what cost?"
- **T5 Measurement** — "Does metric M actually correlate with outcome O that we claim to care about?"

The full statement is **six moves in under 200 words**: context, gap, objective, question, hypothesis, scope. Then score it on **FINER + M** — the criteria *multiply*, so any single 1 is fatal.

COMMON TRAP

Skipping the scope line. Scope says what you *will not* do. It is the only thing that stops a supervisor, an examiner, or you at 2 a.m. from quietly expanding the project.

WORKED EXAMPLE — read it, then delete this whole box

Narrowing trace. Interest: applied NLP → Field: RAG → Topic: faithfulness → Task: answer attribution over institutional policy documents → Problem: retrieval quality is evaluated offline, but what users experience is an unsupported answer inside a latency budget. **Knobs turned:** *population* (university policy and circular documents, not Wikipedia), *setting* (p95 end-to-end under 2 s on one commodity GPU), *metric* (rate of unsupported claims verified against retrieved context, not answer accuracy alone). **Projects that fit:** about one.

[Context] Retrieval-augmented generation is the standard way to ground a language model in an organisation's own documents Lewis et al. [2020], and is widely deployed for institutional question answering. **[Gap]** Published faithfulness evaluations use clean, open-domain corpora and impose no serving constraint; how the unsupported-claim rate behaves on messy institutional documents *under a hard latency budget* — where retrieval depth and reranking must be cut — is reported far less often. **[Objective]** This work measures the trade-off between the latency budget and unsupported-claim rate for a RAG pipeline over 1,200 university policy documents, and identifies which pipeline stage the loss comes from. **[Question]** Does reducing retrieval depth to meet a 2-second p95 end-to-end budget increase the unsupported-claim rate by more than 10 percentage points over an unconstrained pipeline, on a 300-question set built from university policy documents, using the same generator model? **[Hypothesis]** It will; and dropping the cross-encoder reranker will account for more of the increase than reducing top-k alone. **[Scope]** English documents only; one generator model, held fixed; no fine-tuning; no multi-turn dialogue; no user study.

FINER + M (self-scored). F 4 — documents and prototype in hand; annotation is the cost, budgeted. I 4 — any institution deploying a document assistant; our own IT cell has asked. N 4 — type 3 new context plus type 4 new evidence. E 3 — documents are institutional, some contain staff names; needs a redaction step and permission, both in progress. R 5 — if true, deployment guidance changes: keep the reranker, cut top-k. M 4 — unsupported-claim rate is human-verified on a sample. **Lowest row: E at 3.** No row scores 1, but E is the row to fix first.
Repair applied: narrow, then re-anchor. The first version said "improve chatbot accuracy", which named no construct. It now fixes the generator model and measures one thing: unsupported claims.

YOUR WORK

Version 1 • *[date]* • initial statement, after Lecture I.3

Narrowing trace

Interest *[. . .]* → **Field** *[. . .]* → **Topic** *[. . .]* → **Task** *[. . .]* → **Problem** *[. . .]*
Knobs I turned: *[population / task / comparison / metric / setting — name about three]*
How many final-year projects fit my sentence? *[aim for about 1]*

Problem statement (six moves, under 200 words)

[Context] *[why the area matters — 2 lines]*
[Gap] *[what is not known — 1 line]*
[Objective] *[what you will do about it]*
[Question] *[one sentence, in template T_ _]*
[Hypothesis] *[what you expect, stated so it could be wrong]*
[Scope] *[three things you will NOT do]*

FINER + M scorecard

Criterion	Score (1–5)	Evidence — a file, a paper, or a named person
Feasible	*[. . .]*	*[. . .]*
Interesting	*[. . .]*	*[name a specific group who would act differently]*
Novel	*[. . .]*	*[which of the four novelty types, and three papers you stand next to]*
Ethical	*[. . .]*	*[. . .]*
Relevant	*[. . .]*	*[run the "so what" chain until someone acts differently]*
Measurable	*[. . .]*	*[construct → operational definition → metric → threshold]*
Lowest row = my project's real risk		*[. . .]*

Repair applied

[narrow / re-anchor / re-measure / abandon — and what changed]

Revisions (append below, never overwrite)

Version 2 • *[date]* • *[what changed and why]*
[revised statement]

DONE WHEN

About one project fits the sentence. All six moves present, under 200 words, in a named template. Every FINER + M row has a score *and* evidence. No row scores 1. The scope line names three exclusions.

4 Research Questions

Lecture I.6 Update this section after these lectures.

WHAT THIS SECTION IS FOR

One **main research question** (the problem statement's question move, restated cleanly) and **two to four sub-questions**, each answerable by a single experiment or analysis. Questions are interrogative and neutral. Hypotheses are declarative and take a risk. Directional hypotheses ("will drop by more than 10 points") are stronger than non-directional ones ("will differ"), because they can be wrong in a specific way. Commit to what counts as support *before* you run anything.

COMMON TRAP

Questions that cannot be wrong. "How can we improve system performance using modern techniques?" No result could refute it, so no result can support it either. If you cannot state the finding that would make you abandon the hypothesis, it is not a research question.

WORKED EXAMPLE — read it, then delete this whole box

RQ0. Does meeting a 2-second p95 end-to-end latency budget increase the unsupported-claim rate of a RAG pipeline over institutional documents by more than 10 percentage points?

ID	Sub-question	Answered by
RQ1	How does the unsupported-claim rate vary across latency budgets of 1, 2, 4 and ∞ seconds?	E1, budget sweep
RQ2	Which cut costs more — removing the cross-encoder reranker, or reducing top-k from 20 to 5?	E2, ablation
RQ3	Does an automatic faithfulness metric agree with human judgement well enough to replace it on this corpus?	E3, metric-validity study

H1. The unsupported-claim rate at a 2 s budget exceeds the unconstrained rate by more than 10 percentage points. *Refuted if* the increase is under 5 points, or the confidence intervals overlap.
H2. Removing the reranker costs more faithfulness than reducing top-k. *Refuted if* the top-k reduction produces the larger increase.

YOUR WORK

Main research question

RQ0. *[one sentence, the same question as in Section 3]*

Sub-questions

ID	Sub-question	Answered by
RQ1	*[. . .]*	*[which experiment]*
RQ2	*[. . .]*	*[. . .]*
RQ3	*[. . .]*	*[. . .]*

Hypotheses

ID	Hypothesis (directional)	Result that would refute it
H1	*[. . .]*	*[. . .]*
H2	*[. . .]*	*[. . .]*

Revisions

Version 2 • *[date]* • *[what changed after which lecture]*

DONE WHEN

One main question, two to four sub-questions, each mapped to one experiment. Every hypothesis is directional and has a stated refuting result.

5 Research Objectives

Lecture I.6 Update this section after these lectures.

WHAT THIS SECTION IS FOR

One **general objective** (the research question restated as an intention) and **three to five specific objectives** that, if all completed, discharge it.
Use measurable verbs — *measure, compare, quantify, characterise, evaluate, attribute, implement.* Avoid *understand, explore, study, look into*: they name no observable, so they cannot be checked off.
Each specific objective must map to exactly one experiment and produce exactly one artefact — a table, a figure, or a released component. **An objective that produces no artefact is a wish.**

COMMON TRAP

Objectives that are activities. "Study the literature", "learn PyTorch", "set up the cluster" are *tasks* on your plan, not objectives of the research. Also: more objectives than you have weeks.

WORKED EXAMPLE — read it, then delete this whole box

General objective. To quantify the trade-off between end-to-end latency budget and unsupported-claim rate in a retrieval-augmented question answering pipeline over institutional policy documents, and to attribute the loss to specific pipeline stages.

ID	Specific objective	Artefact	Answers
SO1	*Construct* a 300-question evaluation set with gold supporting passages, double-annotated	Released dataset + guideline	—
SO2	*Implement* the pipeline with switchable reranker and top-k, instrumented per stage	Repo + latency breakdown	—
SO3	*Quantify* unsupported-claim rate at four latency budgets, 3 seeds	Figure 2	RQ1
SO4	*Attribute* the loss across reranker removal and top-k reduction	Table 3	RQ2
SO5	*Evaluate* agreement between an automatic faithfulness metric and human labels	Table 4, Cohen's κ	RQ3

Weakest link: *Metric* → *Claim.* If human annotation of "unsupported" is unreliable, every number downstream is unreliable. SO1 therefore includes a written guideline and inter-annotator agreement, and SO5 tests the automatic metric rather than assuming it.

In scope. 1,200 English policy and circular documents; one generator model held fixed; 300 questions. **Delimitations.** No fine-tuning — it would confound the pipeline comparison. No multi-turn dialogue. No live user study. **Limitations.** One institution's document style; one generator model, so conclusions may not transfer to a larger one; annotation by two students, not domain experts.

YOUR WORK

General objective

[one sentence, beginning "To measure / quantify / characterise . . . "]

Specific objectives

ID	Specific objective (measurable verb first)	Artefact produced	Answers
SO1	*[. . .]*	*[Table / Fig.]*	RQ*[. . .]*
SO2	*[. . .]*	*[. . .]*	RQ*[. . .]*
SO3	*[. . .]*	*[. . .]*	RQ*[. . .]*
SO4	*[. . .]*	*[. . .]*	RQ*[. . .]*

Chain check

Trace it left to right and name the missing link, if any:

Problem → Question → Objective → Hypothesis → Method → Data → Metric → Claim

Weakest link in my chain: *[. . .]*

Scope and delimitations

In scope. *[population, tasks, datasets, platforms, time window]*

Delimitations (chosen exclusions, with reasons). *[. . .]*

Limitations (imposed on me, not chosen). *[. . .]*

DONE WHEN

One general objective. Three to five specific objectives, each starting with a measurable verb, each mapped to one artefact and one RQ. Scope, delimitations and limitations are three separate lists.

6 Literature Papers

Lecture III.1–III.3 Update this section after these lectures.

WHAT THIS SECTION IS FOR

Between **10 and 15 papers** you have actually read — not a search-result dump. For each, record enough that you never have to open it again to cite it or argue with it.
Prioritise in this order: (1) the newest survey in your area, (2) the two or three papers your work directly compares against, (3) the papers that established the datasets or metrics you will use, (4) anything that contradicts your hypothesis. **That last category is the one students skip and examiners ask about.**
Add every paper to `references.bib` as you go. Read at minimum the abstract, the last two paragraphs, and the threats-to-validity section.

COMMON TRAP

Citing what you have not read. If you cannot state a paper's baseline, dataset and headline number in one line, you have not read it — you have seen its title. Examiners open one citation at random.

WORKED EXAMPLE — read it, then delete this whole box

P1	`\cite{lewis2020rag}`
Full citation	Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks", NeurIPS 2020.
Problem	Parametric models cannot cite or update their knowledge.
Method	Dense retriever + seq2seq generator, trained end-to-end.
Dataset	Natural Questions, TriviaQA, Wikipedia index.
Metric	Exact match; strong gains over parametric-only baselines.
Limitation	Clean, open-domain Wikipedia corpus; no serving-latency constraint.
Why it matters	Defines my architecture and is the source of both gaps I claim.
Date read	2026-08-09

P2	`\cite{liu2024lostmiddle}`
Full citation	Liu et al., "Lost in the Middle: How Language Models Use Long Contexts", TACL 2024.
Problem	Longer context was assumed to be strictly better.
Method	Controlled position experiments on multi-document QA.
Dataset	Multi-document QA, key-value retrieval.
Metric	Accuracy by gold-passage position — U-shaped curve.
Limitation	Synthetic position control; not measured under a latency budget.
Why it matters	**Contradicts the intuition behind my own H2** — more retrieved passages may not help even when latency permits.
Date read	2026-08-12

(P3–P12 continue in the same form: Karpukhin et al. [2020], Es et al. [2024], Asai et al. [2024], Ji et al. [2023], Thakur et al. [2021], Gao et al. [2023]...)

Search record. Google Scholar, 2026-08-09, `retrieval augmented generation faithfulness evaluation` — 340 hits, 11 kept. ACL Anthology, `RAG hallucination attribution 2023..2026` — 88 hits, 4 kept. **Inclusion:** peer-reviewed or arXiv, 2020 onward, reports a faithfulness or attribution metric. **Exclusion:** no empirical evaluation; purely prompt-engineering reports.

YOUR WORK

For each paper, fill one block. Duplicate the block as many times as needed.

P1	*[cite key, e.g. \cite{smith2024}]*
Full citation	*[authors, title, venue, year]*
Problem addressed	*[one line]*
Method	*[one line]*
Dataset / system	*[. . .]*
Metric + headline number	*[. . .]*
Stated limitation	*[quote it, do not paraphrase]*
Why it matters to me	*[baseline / dataset / metric / contradicts me]*
Date read	*[. . .]*

P2	*[cite key]*
Full citation	*[. . .]*
Problem addressed	*[. . .]*
Method	*[. . .]*
Dataset / system	*[. . .]*
Metric + headline number	*[. . .]*
Stated limitation	*[. . .]*
Why it matters to me	*[. . .]*
Date read	*[. . .]*

[TODO: duplicate the block above for P3 . . . P12]

Search record — so this is reproducible

Database	Date	Query string	Hits / kept
[Scholar]	*[. . .]*	*[. . .]*	*[. . .]*
[arXiv]	*[. . .]*	*[. . .]*	*[. . .]*
[IEEE / ACM DL]	*[. . .]*	*[. . .]*	*[. . .]*

Inclusion criteria. *[. . .]* **Exclusion criteria.** *[. . .]*

DONE WHEN

10–15 papers, each with method, dataset, metric, headline number and a quoted limitation. All are in `references.bib` and compile. At least one paper disagrees with your hypothesis. The search record lets someone repeat your search.

7 Literature Matrix

Lecture III.2–III.4 Update this section after these lectures.

WHAT THIS SECTION IS FOR

The matrix is where a reading list becomes an argument. Organise **by concept, not by paper** — a matrix whose rows are papers and whose columns are "summary" is a list in disguise.

Read the table *down the columns*. The empty cells and the columns where every row says the same thing are your research gap; Section 8 simply writes up what this table already shows.

The table is set landscape so the columns fit. Add rows freely — it breaks across pages.

COMMON TRAP

A matrix with no empty cells. If every cell is filled and every paper agrees, either your columns are too coarse or you have only read papers that support you. Add the column that distinguishes your work, and one paper that contradicts you.

Worked example — delete once you have built your own

Read down the columns. The bold cells in the last row are the ones no published row fills — that is the research gap, and Section 8 simply writes it up.

Paper	Year / venue	Problem addressed	Method	Corpus type	Latency budget	Faithfulness metric	Stated limitation	Relevance to me
Lewis et al. [2020]	2020 NeurIPS	Parametric models cannot cite or update	Dense retriever + seq2seq, end-to-end	Open-domain (Wikipedia)	**none**	Exact match only	No serving constraint	Defines my architecture
Karpukhin et al. [2020]	2020 EMNLP	BM25 misses paraphrase	Dense passage retrieval	Open-domain	**none**	Retrieval recall@k	Retrieval only, no generation	My retriever + a baseline
Es et al. [2024]	2024 EACL	Faithfulness costly to annotate	Reference-free automatic metrics	Open-domain	**none**	Faithfulness, answer relevance	Validated on open-domain only	**The metric I must validate (E3)**
Asai et al. [2024]	2024 ICLR	Retrieval always-on hurts	Self-reflective retrieve/critique	Open-domain	**none**	Factscore, citation precision	Extra inference passes cost latency	Architectural alternative; costs budget
Liu et al. [2024]	2024 TACL	Longer context assumed better	Controlled position experiments	Synthetic multi-doc	n/a	Position-wise accuracy	Synthetic control	**Contradicts my H2 intuition**
Ji et al. [2023]	2023 CSUR	Hallucination taxonomy scattered	Survey	—	—	Survey of metrics	No new evaluation	Terminology and taxonomy
Thakur et al. [2021]	2021 NeurIPS D&B	Retrievers over-tuned to one benchmark	Heterogeneous zero-shot benchmark	18 open datasets	n/a	nDCG@10	No generation, no institutional text	External sanity check for retriever
This notebook	2026	Faithfulness under a serving budget	Budget sweep + component ablation	**Institutional policy docs**	**1/2/4/∞ s**	**Human unsupported-claim rate**	One institution, one model	**The empty column, filled**

Your literature matrix

One row per paper from Section 6. Rename the columns to whatever distinguishes work in your area — the columns are the argument. Keep the last row for your own proposed work. Add rows freely; the table breaks across pages.

Paper	Year / venue	*[column 1]*	*[column 2]*	*[column 3]*	*[column 4]*	*[column 5]*	*[column 6]*	Relevance to me

Paper	**Year / venue**	*[column 1]*	*[column 2]*	*[column 3]*	*[column 4]*	*[column 5]*	*[column 6]*	**Relevance to me**
This notebook	*[. . .]*	*[. . .]*	*[. . .]*	*[. . .]*	*[. . .]*	*[. . .]*	*[. . .]*	*[. . .]*

DONE WHEN

Every paper from Section 6 appears as a row. At least one column is populated for your own proposed work, in the last row. You can point at the specific cells — empty, or uniformly identical — that justify the gap in Section 8.

8 Research Gap

Lecture III.4 Update this section after these lectures.

WHAT THIS SECTION IS FOR

A gap is not "nobody has done exactly this". Name which **kind** of gap it is, and point at the cells in the Section 7 matrix that prove it:

- **Explicit** — the authors state it in Future Work or Limitations. Easiest, and the most crowded.
- **Population** — the method works, but was never tested on your users, language, hardware or region. Best value for a final-year project.
- **Contradiction** — paper A says X helps, paper B says it hurts, nobody reconciled them.
- **Assumption** — everyone assumes Y and nobody tested Y. Hardest to see, highest payoff.
- **Methodological** — the claim exists but the evidence for it is weak: no baseline, one seed, a metric that does not measure the construct.

COMMON TRAP

The crowded gap. You found it in a famous paper's Future Work — so did four hundred other people, and the authors started a year ago. Combine two sources instead: their gap *plus* your population, your constraint, or your hardware.

WORKED EXAMPLE — read it, then delete this whole box

Statement. Faithfulness of RAG systems is now measured routinely, both by human annotation and by automatic metrics Es et al. [2024], and architectural remedies exist Asai et al. [2024]. Every evaluation I found either uses a clean open-domain corpus or imposes no serving constraint. Deployed institutional assistants have both problems at once: messy source documents *and* a latency budget that forces retrieval depth down. Whether the unsupported-claim rate degrades gracefully or sharply under that budget, and which pipeline stage is responsible, is not established.

Field	Answer
Type of gap	**Population** (institutional documents) crossed with **assumption** — everyone assumes the offline retrieval configuration is the deployed one.
Evidence from the matrix	The *latency budget* column reads "none" for every published row; the *corpus type* column reads "open-domain" for all but one.
Closest papers	Es et al. [2024] — same metric, no latency constraint. Liu et al. [2024] — position effects, but synthetic and unconstrained.
Why still open	Benchmark work optimises for comparability, so it removes deployment constraints on purpose.
What changes	**Practice.** Concrete deployment guidance: which component to cut first when you must fit a budget.

Did the gap change my question? Yes. Liu et al. [2024] implies more passages can *hurt*, so my original framing — "latency forces us to retrieve less, therefore quality drops" — was too simple. RQ2 now compares the two cuts against each other instead of assuming both are harmful.

YOUR WORK

Statement of the gap

[3–5 sentences. What is known, what is not known, and why the missing part matters.]

Gap classification

Field	Your answer
Type of gap	*[explicit / population / contradiction / assumption / methodological]*
Evidence from the matrix	*[name the columns and rows — "no row in the X column"]*
Papers that come closest	*[2–3 cite keys, and what each still leaves open]*
Why nobody has closed it yet	*[cost? access? nobody asked? recently became possible?]*
What changes once it is closed	*[knowledge / practice / policy / cost]*

Did the gap change my question?

[If the literature answered your original question, say so and state the revised question. This is a good outcome, not a failure.]

Version 3 • *[date]* • *[question revised after literature study]*

DONE WHEN

The gap is one of the five named types. It is justified by specific cells in the Section 7 matrix, not by assertion. The two closest papers are named, with what each leaves open. Any change to the research question is recorded, not hidden.

9 Proposed Methodology

Lecture I.7, II.1–II.6 Update this section after these lectures.

WHAT THIS SECTION IS FOR

The method follows from the *kind of claim* you want to defend, not from the technique you already know. Pick your primary approach from Lecture I.7 and say why the other three are wrong for this question:

- **Analytical / theoretical** — evidence is a proof or a bound. Holds for all inputs; assumptions carry the argument.
- **Experimental / empirical** — evidence is measured behaviour under controlled variation. Needs a named baseline, held-out data, ≥ 3 seeds.
- **Simulation-based** — evidence is behaviour of a model, when the real system is inaccessible or too large. Validity is inherited from the simulator.
- **Design science / constructive** — evidence is an artefact plus a rigorous evaluation of it. *Problem* → *design* → *demonstration* → *evaluation*.

Most good computing work is a hybrid. Say which parts are which.

COMMON TRAP

Demonstration masquerading as evaluation. Showing that your system runs is not evidence that it is better. If the evaluation cannot come out against you, it is a demo.

WORKED EXAMPLE — read it, then delete this whole box

Primary approach: experimental. **Secondary:** design science — the instrumented, switchable pipeline is a built artefact (SO2), but the claim rests on the measurements.
Why not the others. *Analytical*: there is no tractable model of when a generator will assert something its context does not support. *Simulation*: the phenomenon is the model's behaviour on real text; a simulator would have to contain the thing under study. *Design science alone*: building an assistant and demonstrating it runs would be a project, not research — nothing could come out against me.
Pipeline. 1,200 documents → redaction → chunking (fixed at 512 tokens, 64 overlap) → dense index Karpukhin et al. [2020] → top-k retrieval → optional cross-encoder reranker → prompt assembly → generator (fixed model, temperature 0) → claim extraction → human + automatic support verification. Per-stage latency is logged.
Baselines. (i) *No retrieval* — generator alone, the trivial floor and the reason RAG exists. (ii) *Unconstrained RAG* — top-20 with reranker, no budget: the quality ceiling. (iii) *BM25 retrieval* — the cheap classical comparison that reviewers always ask for.
Validity. *Internal*: generator model, prompt and temperature held fixed; only retrieval configuration varies. *External*: one institution, one model — stated, not claimed away. *Construct*: "unsupported claim" is defined against the retrieved context actually shown to the generator, not against ground truth, because that is what faithfulness means. *Conclusion*: 3 generation seeds, bootstrap CIs, and inter-annotator agreement reported before any rate is reported.

YOUR WORK

Approach

Primary approach. *[analytical / experimental / simulation / design science]*
Secondary approach, if hybrid. *[. . .]*
Why not the others. *[one line each — this is the part examiners probe]*

Pipeline

[Describe the pipeline stage by stage. Then draw it — a figure here is worth a page of prose. Use `\includegraphics` or a TikZ diagram.]

Replace this box with your pipeline diagram
(`figures/pipeline.pdf` or a TikZ picture)

Baselines — what you are comparing against

Baseline	Why this is the right comparison	Source / cite key
[trivial baseline]	*[establishes the floor]*	*[. . .]*
[standard practice]	*[what people do today]*	*[. . .]*
[published SOTA]	*[the number to beat]*	*[. . .]*

Validity

Threat	How I address it
Internal	*[did the thing you changed cause the effect?]*
External	*[does it hold outside your sample?]*
Construct	*[did you measure the thing you named?]*
Conclusion	*[do the statistics support the strength of the claim?]*

DONE WHEN

Primary approach named, with one line on why each of the other three is wrong here. A pipeline figure exists. At least two baselines, one of them trivial. All four validity threats have a stated mitigation.

10 Dataset / Data Source

Lecture I.8 Update this section after these lectures.

WHAT THIS SECTION IS FOR

Name the exact data, systems, users or experimental apparatus you need — and prove you can get it. **Confirm access in writing in week one.** A beautiful problem whose data sits behind an eight-month approval is not a problem you can do, and narrowing will not fix it: a smaller slice of inaccessible data is still inaccessible.
Record licence and provenance for every source. "Publicly visible" is not "public", and "public" is not "yours to use".

COMMON TRAP

Data leakage, designed in on day one. Separate your held-out test set *before* any preprocessing, tuning or modelling, and touch it once. Watch for preprocessing fitted on the full set, temporal shuffling of time-ordered data, and near-duplicates across splits.

WORKED EXAMPLE — read it, then delete this whole box

Source	What it gives / size	Licence	Access confirmed
University policy documents	1,200 PDFs, circulars and handbooks	Institutional; internal-use permission	2026-08-11, written approval from the Registrar's office, redaction required
Question set (built by me)	300 questions with gold supporting passages	CC-BY-4.0 on release, post-redaction	Under construction
BEIR Thakur et al. [2021]	External sanity check for the retriever	Apache-2.0	Downloaded

Building the question set. Questions are drawn from real queries in the IT helpdesk log (de-identified) plus questions written by two annotators while reading the corpus. **Sampling:** stratified by document type (academic policy / finance / hostel / examination) so no category dominates. **Size:** 300, chosen so that a 10-point difference in rate is detectable at 80% power. **Annotation:** two annotators, written guideline, every item double-annotated, disagreements adjudicated by a third pass; Cohen's κ reported before any result.

Splits. 60 questions form a development set used for prompt and threshold decisions. 240 are the evaluation set, sealed until all configurations are frozen and touched exactly once.
Leakage audit. No question is written after seeing model output. Prompt and chunking decisions use the development set only. The evaluation set is opened once, on a recorded date.
Fallback. If redaction clearance is delayed, the same design runs on a public institutional corpus (a university's published handbooks) at the cost of realism in the messiness of the documents — reported as a limitation.

YOUR WORK

Sources

Source	What it gives me / size	Licence	Access confirmed?
[. . .]	*[. . .]*	*[. . .]*	*[date + how]*
[. . .]	*[. . .]*	*[. . .]*	*[. . .]*
[. . .]	*[. . .]*	*[. . .]*	*[. . .]*

If I have to build the data myself

Collection procedure. *[detailed enough that a stranger could repeat it]*

Sampling method. *[random / stratified / purposive / convenience — and declare convenience honestly]*

Target size, and why that size. *[. . .]*

Annotation. *[who labels, against what guideline, how agreement is measured]*

Splits

Split	Size	How separated (and when)
Train	*[. . .]*	*[. . .]*
Validation	*[. . .]*	*[. . .]*
Test (touched once)	*[. . .]*	*[separated before any preprocessing]*

Leakage audit

[Preprocessing fitted on train only? Time order preserved? Duplicates across splits checked? Test set used exactly once?]

Fallback plan

[If the primary source falls through, what do I use instead — and what does that cost the claim?]

DONE WHEN

Every source has a licence and a confirmed access route with a date. Splits are defined, with the test set separated before preprocessing. A leakage audit is written. A fallback source exists.

11 Experimental Plan

Lecture I.8, II.7 Update this section after these lectures.

WHAT THIS SECTION IS FOR

The experiment that would settle the question, described so precisely that someone else could run it and get your numbers.
State independent variables (what you change), dependent variables (what you measure), and controlled variables (what you hold fixed). Run **at least three seeds** — a single run is an anecdote — and report mean with standard deviation.
Then **cost it before you commit**. Scope creep is multiplicative, not additive: one extra baseline multiplies through every dataset, seed and ablation.

COMMON TRAP

An unfair tuning budget. Tuning your method for a week and running the baseline once, out of the box, is not a comparison. Give the baseline the same budget, and say what budget that was.

WORKED EXAMPLE — read it, then delete this whole box

Variables. *Independent*: latency budget (1, 2, 4, ∞ s), reranker (on/off), top-k ($\in \{5, 20\}$). *Dependent*: unsupported-claim rate, answer accuracy, p95 end-to-end latency. *Controlled*: generator model and version, temperature 0, prompt template, chunking, index build, hardware, seed set $\{1, 2, 3\}$.

ID	Name	Varies / answers	Output
E1	Budget sweep	4 budgets × 3 seeds — RQ1	Fig. 2
E2	Component ablation	reranker on/off × top-k — RQ2	Table 3
E3	Metric validity	automatic faithfulness vs. human labels — RQ3	Table 4

Cost model, run 2026-08-14. `configs=8, seeds=3, questions=240, gen_time=1.8s` $\Rightarrow 8 \times 3 \times 240 \times 1.8\,\mathrm{s} \approx 2.9$ GPU-hours — trivial. **The real budget is human:** 240 questions × 8 configs is far too much to annotate, so support verification is done on a stratified sample of 60 questions per configuration ($8 \times 60 \times 2$ annotators $\times$ 2.5 min $\approx$ 40 person-hours), with the automatic metric (validated in E3) covering the rest. **This is the constraint that shaped the design**, and it is why E3 exists at all.
Reproducibility. Repo `github.com/<me>/rag-latency-faithfulness`; model version and prompt hash logged per run; `make figures` regenerates every plot; annotation guideline and adjudication log released with the dataset.
Timeline. W1–2 corpus + redaction + prototype (crude end-to-end result by day 6); W3–5 question set and annotation guideline; W6–8 annotation; W9–11 E1/E2; W12 E3; W13–14 write-up.

YOUR WORK

Variables

Type	Variables
Independent (changed)	*[. . .]*
Dependent (measured)	*[. . .]*
Controlled (held fixed)	*[hardware, versions, seeds, load, temperature]*

Experiments

ID	Name	What it varies, and which RQ it answers	Output
E1	*[main comparison]*	*[. . .]*	*[Table 1]*
E2	*[ablation]*	*[. . .]*	*[Fig. 2]*
E3	*[sensitivity / stress]*	*[. . .]*	*[. . .]*

Cost model — run this before you commit

```
methods   = 3      # yours + how many baselines?
datasets  = 2
seeds     = 3      # minimum for any honest claim
ablations = 4
hours_per_run = 1.5    # measured on ONE short pilot run, never guessed

runs  = methods * datasets * seeds * (1 + ablations)
total = runs * hours_per_run
print(runs, "runs |", total, "compute-hours")

BUDGET = 60        # what you can actually book, not what exists
assert total <= BUDGET, f"need {total} h, have {BUDGET} -- narrow a knob"
```

My numbers. runs = *[. . .]*, total = *[. . .]* hours, budget = *[. . .]* hours.
If the assertion fails, the knob I turned: *[. . .]*

Reproducibility

[Repository link, pinned dependency versions, seeds recorded, one command that regenerates every table and figure, hardware and runtime logged.]

Timeline

Weeks	Activity	Deliverable
[1–2]	*[data access + pilot run]*	*[crude end-to-end result]*
[3–5]	*[reproduce baseline]*	*[. . .]*
[6–9]	*[main experiments]*	*[. . .]*
[10–12]	*[ablations + repeats]*	*[. . .]*
[13–14]	*[write-up + figures]*	*[. . .]*

DONE WHEN

Variables classified into three lists. Every experiment maps to an RQ and names its output artefact. The cost model has been run with your own numbers and fits your budget. A crude end-to-end result exists, or is scheduled, in week one.

12 Evaluation Metrics

Lecture I.4, II.8 Update this section after these lectures.

WHAT THIS SECTION IS FOR

Take every fuzzy word in your problem statement — “better”, “efficient”, “usable”, “fair” — and push it through the chain:

Construct → Operational definition → Metric → Threshold vs. a named baseline

Then interrogate each metric twice. The **validity gap**: does the metric still mean the construct? The **gaming gap**: can the metric be won without the construct? *Commits per week* measures activity, not value — and anyone who knows the metric can split one change into fifteen commits.
Report a **primary** metric (the claim rests on it), **secondary** metrics, and **guardrail** metrics that must not get worse.

COMMON TRAP

Metric shopping. Changing the metric or dataset until the result looks good, after seeing the results. This is a research-integrity violation, not a tuning step — and Unit III will name it. Fix the metric and threshold before the first run.

WORKED EXAMPLE — read it, then delete this whole box

Construct	Operational definition	Metric	Threshold
“faithful”	every atomic claim in the answer is entailed by a passage actually shown to the generator	unsupported-claim rate (% of claims)	$\leq$ unconstrained rate + 5 pp
“useful”	answer contains the gold fact	answer accuracy (human-judged)	$\geq$ 80% of unconstrained
“fast”	request receipt to last response token	p95 end-to-end latency	$\leq$ 2 000 ms
“reliable annotation”	two annotators agree on support	Cohen’s κ	$\geq$ 0.7 before any rate is reported

Primary: unsupported-claim rate. **Secondary:** answer accuracy. **Guardrails:** p95 latency must stay within budget, and the abstention rate must not rise — otherwise the system wins by refusing to answer.

Metric	Validity gap	Gaming gap
Unsupported-claim rate	A short, vague answer has few claims and so few unsupported ones — it measures caution as much as faithfulness	Answer “I don’t know” every time and score perfectly — hence the abstention-rate guardrail and the accuracy secondary
Automatic faithfulness score	Correlates with human judgement on open-domain text; unvalidated on policy prose full of cross-references	Optimise phrasing toward whatever the judge model rewards — hence E3 validates it before it is used

Statistics. 3 generation seeds; rates reported with bootstrap 95% CI; McNemar’s test for paired per-question comparisons between configurations; Holm correction across the four budgets; Cohen’s κ for annotation agreement, reported first.
Success criterion, fixed 2026-08-14 before the evaluation set was opened. H1 is supported if and only if the unsupported-claim rate at the 2 s budget exceeds the unconstrained rate by more

than 10 percentage points, with non-overlapping bootstrap CIs, on the sealed 240-question set.

YOUR WORK

Construct to threshold

Construct	Operational definition	Metric	Threshold
[. . .]	*[. . .]*	*[. . .]*	*[vs. named baseline]*
[. . .]	*[. . .]*	*[. . .]*	*[. . .]*
[. . .]	*[. . .]*	*[. . .]*	*[. . .]*

Metric roles

Primary (the claim rests on this one): *[. . .]*
Secondary: *[. . .]*
Guardrails (must not get worse): *[. . .]*

Validity and gaming

Metric	Validity gap — where it stops meaning the construct	Gaming gap — how it could be won dishonestly
[. . .]	*[. . .]*	*[. . .]*
[. . .]	*[. . .]*	*[. . .]*

Statistical treatment

Seeds / repetitions: *[≥ 3]* **Reported as:** *[mean $\pm$ SD, 95% CI]*
Significance test and why it is appropriate: *[. . .]*
Effect size measure: *[. . .]*
Multiple-comparison correction, if testing many configurations: *[. . .]*

Success criterion, fixed in advance

["I will claim success if and only if . . . " — write it before the first run, and do not edit it afterwards.]

DONE WHEN

Every fuzzy word in the problem statement has a row. One primary metric is named. Each metric has both a validity gap and a gaming gap written down. The success criterion is fixed and dated before the first experiment.

13 Ethical Considerations

Lecture III.5–III.8 Update this section after these lectures.

WHAT THIS SECTION IS FOR

Ethics is a gate, not a form — and you cannot narrow your way past it. Work through the five gates from Lecture I.4, then the integrity items from Unit III.

G1. Human subjects or their data? Consent, and ethics-committee approval.

G2. Data obtained lawfully? Licence, terms of service, `robots.txt`.

G3. Could the result be misused? Dual use; staged or withheld release.

G4. Could it harm one group more than another? Disaggregated evaluation.

G5. Is it honestly reported? Negative results, limitations, no metric shopping.

If your work touches no human data at all, say so explicitly and justify it — "not applicable" with a reason is a valid, and expected, answer.

COMMON TRAP

"It is public, so it is fine." Visible is not consented, and public is not licensed. Scraping a site that forbids it in its terms of service is a violation whether or not anyone notices.

WORKED EXAMPLE — read it, then delete this whole box

	Gate	Applies	Answer / mitigation
G1	Human subjects or their data	**Yes**	Helpdesk queries are user-generated. Used only after de-identification; no user is contacted; no demographic attribute is retained. Annotators are two consenting classmates, credited.
G2	Lawful acquisition	**Yes**	Written permission from the Registrar's office, dated 2026-08-11, for internal research use with redaction. Public release covers only the redacted question set, not the source PDFs.
G3	Misuse / dual use	**Moderate**	An attribution-failure catalogue is effectively a list of prompts that make the assistant lie. Failure cases are released in aggregate form, not as a ready-to-use prompt set.
G4	Unequal harm	**Yes**	Wrong answers about fees, scholarships or hostel policy fall hardest on students least able to absorb the error. Results are stratified by document category, and the finance/scholarship stratum is reported separately even if it is small.
G5	Honest reporting	**Always**	Success criterion fixed before the sealed set was opened; κ reported before rates; a null result is reported as the headline if it occurs.

Privacy and security. Staff names, phone numbers, employee IDs and signatures are removed by an automated redaction pass with a manual check on a 10% sample. Documents stay on an encrypted institutional drive; no document is sent to any third-party API — the generator runs locally, which is also why the latency budget is a real constraint rather than a hypothetical one.
Bias and fairness. Known bias: the corpus over-represents academic policy and under-represents hostel and accessibility circulars, so error rates on the smaller strata will have wide intervals —

reported with the intervals, not hidden by pooling.
Integrity. Turnitin check before submission. Two annotators acknowledged by name with their consent. **AI assistance:** used to draft the redaction regex list and to reformat tables; every annotation guideline decision, every reported number and every citation was made and verified by me. Negative result committed to in advance.

YOUR WORK

The five gates

	Gate	Applies?	My answer / mitigation
G1	Human subjects or their data	*[Y/N]*	*[. . .]*
G2	Lawful acquisition (licence, ToS)	*[Y/N]*	*[. . .]*
G3	Misuse / dual use	*[Y/N]*	*[. . .]*
G4	Unequal harm across groups	*[Y/N]*	*[. . .]*
G5	Honest reporting	**Always**	*[. . .]*

Privacy and security

Personal data involved: *[what fields, whose]*
Anonymisation / de-identification: *[. . .]*
Storage, access control and retention: *[where it lives, who can read it, when it is deleted]*

Bias and fairness

Groups I will report results separately for: *[. . .]*
Known bias in my data source: *[. . .]*
What I will do if a disparity appears: *[. . .]*

Research integrity

- **Plagiarism.** *[quoting and paraphrasing policy; similarity check tool and date]*
- **Authorship and credit.** *[who contributed what]*
- **AI-assistance disclosure.** *[which tools, for what — writing, coding, analysis — and what you verified yourself]*
- **Negative results.** *[committed to reporting them]*
- **Conflicts of interest / funding.** *[. . .]*

Declaration

I confirm that the work recorded in this notebook is my own, that all sources are cited, and that any assistance is disclosed above.

Signature: ______________________ Date: ______________

DONE WHEN

All five gates answered, including the ones marked not-applicable, each with a reason. Privacy, bias and integrity subsections are complete. The AI-assistance disclosure is specific about what was verified. The declaration is signed.

14 Research Proposal

Lecture IV.1–IV.5 Update this section after these lectures.

WHAT THIS SECTION IS FOR

The proposal is not new writing — it is Sections 1–13 assembled into a document someone outside the course could evaluate. If a subsection here needs material that is not already in your notebook, that is the notebook telling you what to work on next.
Target **8–12 pages** excluding references. Write for a reader who knows computer science but not your subfield.

COMMON TRAP

Relevance theatre. "This will help society and contribute to the nation's growth." A sentence that fits every project fits none. If your significance paragraph would survive being pasted into someone else's proposal, it is not a significance paragraph.

WORKED EXAMPLE — read it, then delete this whole box

Title. *What Does a Latency Budget Cost You in Faithfulness? Measuring Unsupported Claims in Retrieval-Augmented Question Answering over Institutional Documents*

Abstract (excerpt). Retrieval-augmented generation is the standard way to ground a language model in an organisation's own documents Lewis et al. [2020], and faithfulness is now routinely measured Es et al. [2024]. Published evaluations, however, use clean open-domain corpora and impose no serving constraint, while deployed assistants must answer inside a latency budget that forces retrieval depth down. This work measures the unsupported-claim rate of a locally-served RAG pipeline over 1,200 university policy documents at budgets of 1, 2, 4 and unlimited seconds, on a sealed 240-question double-annotated evaluation set, and attributes any degradation to reranker removal versus reduced top-k. We hypothesise a rise of more than 10 percentage points at a 2-second budget, driven mainly by reranker removal. A separate study validates an automatic faithfulness metric against human judgement on this corpus before it is used . . .

Expected contribution: "*new evidence* plus a released, redacted benchmark: concrete guidance on which pipeline component to cut first under a budget, and the first faithfulness evaluation on institutional policy prose rather than encyclopaedic text."

YOUR WORK

Title

[Specific enough that a reader knows the population, the task and the setting. Twelve to eighteen words.]

Abstract (150–250 words)

[Problem, gap, what you will do, how you will evaluate it, expected contribution. Write this last.]

1. Introduction and motivation

[From Sections 1 and 3. End with the "so what" chain from Section 3.]

2. Problem statement and research questions

[From Sections 3 and 4, verbatim.]

3. Literature review

[From Sections 6 and 7 — written as prose organised by concept, with the matrix as a table. Not a list of paper summaries.]

4. Research gap

[From Section 8.]

5. Objectives

[From Section 5.]

6. Proposed methodology

[From Section 9, including the pipeline figure and the baselines.]

7. Data and experimental plan

[From Sections 10 and 11, including the timeline.]

8. Evaluation

[From Section 12, including the success criterion fixed in advance.]

9. Ethical considerations

[From Section 13.]

10. Expected contribution and limitations

[What will be true that is not true today — in knowledge, practice, policy or cost. Then the limitations, stated by you before anyone else states them.]

11. Timeline and deliverables

[Gantt-style table or list.]

References

[Generated automatically from `references.bib`.]

Self-assessment before submission

Question	Yes/No
Could a stranger run my experiment from Section 7 alone?	[. . .]
Does every claim in Section 10 trace to a metric in Section 8?	[. . .]
Is there a result that would make me abandon my hypothesis, and is it stated?	[. . .]
Have I cited at least one paper that disagrees with me?	[. . .]
Does my significance paragraph fail the paste-into-someone-else's-proposal test?	[. . .]

DONE WHEN

Every subsection is drawn from an earlier notebook section, not written fresh. 8–12 pages. The self-assessment table is complete and honest. References compile from `references.bib`.

15 Final Presentation

Lecture IV.5 Update this section after these lectures.

WHAT THIS SECTION IS FOR

Record the **plan** for your defence and, afterwards, what actually happened. The second part is the one that is marked — it shows you listened.
Suggested arc for 10–12 slides: hook (the concrete failure or cost that motivates this) → problem statement → what is known → the gap → your question and hypothesis → method → data → evaluation plan → expected contribution → limitations and risks.
Rehearse the questions you hope nobody asks. They are the ones that get asked.

COMMON TRAP

Defending instead of thinking. If a question exposes a real weakness, say so, say what you would do about it, and move on. Examiners mark the quality of the answer, not the absence of the weakness.

WORKED EXAMPLE — read it, then delete this whole box

Slide 2, the hook. A screenshot of my own prototype confidently stating a fee deadline that does not exist in any document it retrieved. Underneath, one line: *it was not wrong because the model was small. It was wrong because we cut retrieval to hit 2 seconds.*

Two of the anticipated questions, answered.

"Why not just use a bigger model?" — Because the question is about the retrieval configuration, and a bigger model changes both the latency budget and the faithfulness at once. The model is held fixed on purpose; varying it is a different study, and I say so in the scope.

"Does your metric measure what you claim?" — Partly, and I measure that too. The unsupported-claim rate rewards short cautious answers, so I report abstention rate as a guardrail and accuracy as a secondary. RQ3 exists because I did not want to assume the automatic metric transfers to policy prose.

After the defence (filled in on 2026-11-27). *Example of what this subsection should look like:* "I was asked why $\kappa \geq 0.7$ rather than 0.8, and did not have a real answer — I had taken 0.7 from a blog post. I have since traced the convention to Landis and Koch's 'substantial' threshold and cited it properly, and noted that our adjudication pass makes the raw κ a conservative floor."

YOUR WORK

Slide plan

#	Slide	The one thing this slide must land	Min
1	Title	*[. . .]*	0.5
2	Hook	*[a concrete failure, cost or number]*	1
3	Problem statement	*[. . .]*	1
4	What is known	*[. . .]*	1.5
5	The gap	*[. . .]*	1
6	Question + hypothesis	*[. . .]*	1
7	Method	*[. . .]*	1.5
8	Data	*[. . .]*	1
9	Evaluation	*[. . .]*	1.5
10	Expected contribution	*[. . .]*	1
11	Limitations and risks	*[. . .]*	1

Anticipated questions — and my honest answers

Question I expect	My answer
Why this baseline and not a stronger one?	*[. . .]*
What result would prove you wrong?	*[. . .]*
Why is this novel — which of the four types?	*[. . .]*
Can you actually get this data?	*[. . .]*
Does your metric measure what you claim?	*[. . .]*
[the one you hope nobody asks]	*[. . .]*

After the defence — what actually happened

Date: *[. . .]*

Questions I was actually asked:

1. *[. . .]*
2. *[. . .]*
3. *[. . .]*

The question I answered worst, and what I now know: *[. . .]*

What I would change about the project: *[. . .]*

What I would change about how I worked: *[. . .]*

Closing reflection

Answer the question this notebook exists to answer, in one paragraph:

What is my research problem, why does it matter, what has already been done, what is still missing, how will I investigate it, how will I evaluate it, and what contribution do I expect to make?

[. . .]

DONE WHEN

Slide plan totals 10–12 minutes. Five or more anticipated questions have written answers. The post-defence section is filled in after the defence, including the question you answered worst. The closing reflection answers the course question in one paragraph.

References

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